Sine rule

1
a False
b False
c True
d True

2 No, because the sine rule requires one angle to find another angle.

3 Yes, because we can find the opposite angle to c using the angle sum of a triangle. Then we can find the other two sides using sine rule.

4 Yes, because we can find the third angle using the angle sum of a triangle. Then we can find the other two sides using sine rule.

5 No, because we do not know the side opposite $\angle C$, so we do not have an angle-side pair to use to find any other angles.

6
a $\sin A = \frac{x}{b}$
b $x = a \sin B$
c $\frac{\sin A}{a} = \frac{\sin B}{b}$

Unknown sides

7 $h = 86.52$

8
a $a = 10.5$
b $a = 6.9$
c $x = 13.1$
d $x = 12.9$
e $y = 8.7$
f $y = 4.4$

9
a $a = 16.66$
b $a = 16.66$

10
a 7.2 mm
b 19.4 mm

11
a $HK = 733.17 \text{ km}$
b $KJ = 444.55 \text{ km}$

Unknown angles

12

a $x = 30.4^\circ$ b $x = 32.7^\circ$ c $x = 26.0^\circ$ d $x = 43.2^\circ$

13 $14^\circ 47'$

14

a $A = 75.78^\circ$ b $b = 3.152\text{m}$ c $c = 5.723\text{m}$

15

a $\angle OBA = 39^\circ$ b $k = 21.76$

16

a $x = 140.42^\circ$ b $x = 133.13^\circ$ c $x = 149.12^\circ$

17 $\theta = 140^\circ 40'$

Ambiguous case

18

a $8 < \text{Length} < 10$ b $\text{Length} = 8, \text{Length} \geq 10$

c $\text{Length} < 8$

19

a $\theta = 26^\circ$ b 8.7385 c 9.1247

d No, according to sine rule, $\frac{8.4}{\sin 106^\circ} = \frac{4.0}{\sin 26^\circ}$. Since they are not equal, this triangle is not possible.

20

a No b No c Yes d No

21

a 0 b 1 c 1 d 2

22

a No

b No

c Yes

d No

e Yes

f Yes

g No

h Yes

i Yes

23

a $x = 41$

b 139

24

a

i It does exist

ii It could be either acute or obtuse.

b

i Does not exist.

ii No solution

25

a $x = 72.53^{\circ}$

b $x = 107.47^{\circ}$

26 $3.5 < BC < 5$

Applications

27

a $\angle WST = 136^{\circ}$

b $SW = 234 \text{ km}$

c $h = 102.58 \text{ km}$

28

a $\angle ABC = 21.89^{\circ}$

b $\angle BCA = 123.11^{\circ}$

c $AB = 29.2 \text{ km}$

29

a $\angle A = 38^{\circ}$

b $c = 22.71 \text{ m}$

c $b = 22.71 \text{ m}$

d 1.9 min

30

a $a = 74^{\circ}$

b $d = 28.35 \text{ m}$

c $h = 16.66 \text{ m}$

31

a $\angle ADB = 23^{\circ}$

b $a = 35.65 \text{ m}$

c $h = 20.4 \text{ m}$

32

a $\angle ACB = 102^{\circ}$

b $b = 226.55 \text{ m}$

c $h = 117 \text{ m}$

33

a $BC = 307.3 \text{ m}$ **b** $AC = 335.5 \text{ m}$ **c** 41 m/min